Experiment No: 6

Title: Write a Python Program for the Vacuum Cleaner Problem

Aim

To develop a Python program to simulate the **Vacuum Cleaner Problem** using a simple reflex agent that cleans all dirty rooms and displays the final room status.

Objective

To implement a Vacuum Cleaner agent in Python that senses the condition of each room and performs the cleaning operation until all rooms become clean.

Algorithm

- **Step 1:** Start the program.
- **Step 2:** Define the rooms and their initial status (Dirty/Clean).
- **Step 3:** Read the status of each room.
- **Step 4:** Check whether the current room is dirty.
- **Step 5:** If the room is dirty, clean the room.
- **Step 6:** Move to the next room and repeat the process.
- **Step 7:** Display the final status of all rooms.
- **Step 8:** Stop the program.

Program

```
# Vacuum Cleaner Problem using DFS


rooms = {}


n = int(input("Enter the number of rooms: "))


for i in range(n):
    room = input(f"Enter Room Name {i+1}: ")
    status = input(f"Enter Status of {room} (Dirty/Clean): ").capitalize()
```

```python
    rooms[room] = status

print("\nInitial Room Status")
print(rooms)

visited = set()

def dfs(room):
    if room in visited:
        return

    visited.add(room)

    if rooms[room] == "Dirty":
        print(f"{room} is Dirty -> Cleaning...")
        rooms[room] = "Clean"
    else:
        print(f"{room} is already Clean.")

print("\nCleaning Process")

for room in rooms:
    if room not in visited:
        dfs(room)

print("\nFinal Room Status")
print(rooms)
```

Sample Input

Enter the number of rooms: 2

Enter Room Name 1: A

Enter Status of A (Dirty/Clean): Dirty

Enter Room Name 2: B

Enter Status of B (Dirty/Clean): Dirty

Sample Output

```
Enter the number of rooms: 2
Enter Room Name 1: A
Enter Status of A (Dirty/Clean): Dirty
Enter Room Name 2: B
Enter Status of B (Dirty/Clean): Dirty

Initial Room Status
{'A': 'Dirty', 'B': 'Dirty'}

Cleaning Process
A is Dirty -> Cleaning...
B is Dirty -> Cleaning...

Final Room Status
{'A': 'Clean', 'B': 'Clean'}
```

Result

Thus, the Python program for the **Vacuum Cleaner Problem** was executed successfully, and all dirty rooms were cleaned by the Vacuum Cleaner agent.